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New Technique for Solving Harmonic Oscillator System in Momentum Space

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Abstract. Laplace transformation has been used for solving Schrödinger equation in momentum space without using integral equation (Fourier transformation) because it is much harder and more complicated. Furthermore, energy spectra and particle distributions have been demonstrated by using Schrödinger differential equation without ignoring dimensions or units. In addition, a maple program has also been considered to solve the method since it can help to pave a way for other models to work in quantum mechanics.

1. INTRODUCTION

Harmonic motion is one of the most important examples of motion in physics. Any vibration with a restoring force equals to Hooke's law is generally caused by a simple harmonic oscillator. Harmonic motion plays a fundamental role as a stepping stone in more rigorous applications. Harmonic oscillators arise in various fields such as classical mechanics, electronic engineering, experimental and quantum physics, etc. In quantum mechanics there is no single-particle system that is at the same time as simple and important as the harmonic oscillator. The literature is rich with several approaches to deal with such a fundamental problem. Standard techniques are based on the celebrated Sommerfeld polynomial approach [1, 2], on the operatorial algebra of Fock states [3–5], or even on Feynman's path integral formulation [6].

Furthermore, interesting revisitations of the quantum harmonic oscillator problem, based on Fourier transform technique, are proposed in recent times [7–12]. The majority of researchers have been using Harmonic Oscillator in position representation. On this other hand, the momentum representation is seldom used in quantum mechanics for Harmonic Oscillator potential.

However, the problem of the hydrogen atom has played a central role in the development of quantum mechanics. Schrödinger solved his equation and found the wave function of the coordinate representation. The problem in momentum space has been reformulated by Fock [13] which led to an integral form of the Schrödinger equation. This equation is solved by projecting the three-dimensional momentum space onto the surface of a four-dimensional sphere and the eigenfunctions are then expanded in terms of spherical harmonics. Despite the importance of Fock's work and the interest of many authors [14–20] to study the wave function in momentum space, it must not hide that the direct calculation of Fourier transform of the wave function of coordinates is up till now undone and our aim in this work is to fill this gap.

We know the momentum representation is seldom used in quantum mechanics courses. Some students are thence surprised by the change in viewpoint when, in doing advanced work, they have to use the momentum rather than the coordinate representation. However, we take a model for vibrating atoms in crystal lattices. Accordingly, we have to use momentum space for solving this model, because when waves are assigned to particles in motion indeterminacy arises automatically in the formalism because an electron wave of definite frequency is not localized. Moreover, in this research, we show that a power of Laplace transform scheme can be used to solve the Schrodinger equations in closed form for all known solvable models especially for harmonic oscillator potential. The work demonstrates how to apply the Laplace transform to differential equations with non-constant coefficients, which is

useful in many branches of physics in addition to quantum mechanics. In section 2 we show that two theorems to solve differential equations with variable coefficient. Any periodic back and forth motion produced by a restoring force that is directly proportional to the displacement and in the opposite direction. The displacement centers around an equilibrium position $F \propto x$. Thus, any vibration with a restoring force equals to Hooke's law is generally caused by a simple harmonic oscillator. The potential for the harmonic oscillator is the natural solution for every potential with small oscillations at the minimum. Almost all potentials in nature have small oscillations at the minimum, including many systems studied in quantum mechanics [21].

Consider the one-dimensional motion of a mass point m under the action of an elastic restoring force having spring constant $k = m\omega^2$, with ω being the angular frequency (2π times the harmonic motion frequency), so that it is described through the following potential energy $\hat{V}(x)$:

$$\hat{V}(x) = \frac{1}{2} m\omega^2 \hat{x}^2 \quad (1)$$

To work in momentum space, we use the result

$$\hat{x} = i\hbar \frac{d}{dp} \quad (2)$$

Then equation (1) becomes

$$\hat{V}(p) = \frac{1}{2} m\omega^2 \left(-\hbar^2 \frac{d^2}{dp^2} \right) \quad (3)$$

Once equation (1) is substituted into the one-dimensional stationary-state Schrödinger equation, the following second-order ordinary differential equation for the wave function $\Psi(p)$ describing the stationary state corresponding to the energy E is obtained: [22]

$$\frac{\hat{p}^2}{2m} \Psi(p) + \frac{1}{2} m\omega^2 \hat{x}^2 \Psi(p) = E \Psi(p) \quad (4)$$

By substituting equation (2) in equation (1.4) the Schrodinger equation in momentum space can be written

$$\frac{p^2}{2m} \Psi(p) - \frac{\hbar^2}{2} m\omega^2 \frac{d^2}{dp^2} \Psi(p) = E \Psi(p) \quad (5)$$

2. LAPLACE INTEGRAL TRANSFORMATION

The Laplace transform is one of the most powerful methods, which can be used to solve constant coefficient differential equations. For example, those equations are representing control systems or linear electric circuits. This method has been considered to analyze linear systems in many engineering fields. Laplace transform uses the similar approach as the Fourier transform, which is a reeducation of differential equations with constant coefficients and transforming to algebraic equations owing to easily finding the solutions. Their difference is that the Laplace transform includes the initial conditions in the transform. This is a desirable method to clarify the initial values' effects on the transient response of a dynamical system. In addition, it can be used for exponential order functions while the Fourier transform is restricted to functions of square-integral. Nevertheless, the power expansion method is considered in numerous books for non-constant coefficient differential equations like Schrodinger equation. The Laplace transform with an equation of having a coefficient x^n is an n th order differential equation. If $n > 1$ the order of the equation is not reduced. Thus, it can be argued that the Laplace transform might not be useful in general for non-constant coefficients of differential equations. The power expansion method has however a drawback, which is not yield closed-form expressions for the solutions. In terms of mathematical analysis, closed-form solutions are advantageous [23].

The Laplace transformation of a real function $f(x)$ is defined as

$$F(s) = L[f(x)] = \int_0^{\infty} f(x) e^{-sx} dx, \quad (6)$$

And the inverse Laplace transform is [24, 25].

$$f(x) = L^{-1}[F(s)] = \lim_{y \rightarrow \infty} \frac{1}{2\pi i} \int_{c-iy}^{c+iy} F(s) e^{sx} ds, \quad (7)$$

In which the integration is taken in the complex plane of $z = x + iy$ along the contour $x = c$ selected to the right of all the poles of the function $F(s)$. Equation (7) can also be written as [26, 27].

$$f(x) = L^{-1}[F(s)] = \frac{1}{2\pi i} \int_{\Gamma} F(s) e^{sx} ds, \quad (8)$$

Using Laplace transform for solving differential equations has two theorems:

Theorem 1. If $f^{(k)}(x)$ are continuous for $x \geq 0$ and of exponential order, i.e. $|f^{(k)}(x)| \leq M e^{cx}$ as $x \rightarrow \infty$, with $k=0,1,2,\dots,n$, then for $s > c$, $L[f(x)] = F(s)$ exists and one has [28].

$$L[f^{(n)}(x)] = s^n F(s) - s^{n-1} f(0) - \dots - f^{(n-1)}(0) \quad (9)$$

Theorem 2. If $f(x)$ is continuous for $x \geq 0$ and $|f(x)| \leq M e^{cx}$ as $x \rightarrow \infty$, then for $s > c$, $L[f(x)] = F(s)$ exists and one has [29].

$$L[f^{(n)}(x)] = (-1)^n F^{(n)}(s) \quad (10)$$

and the inverse Laplace transformation

$$L^{-1}[F^{(n)}(s)] = (-x)^n f(x) \quad (11)$$

The initial condition can be ignored because all the Schrodinger equation considered in this research is linear and homogeneous. The solution $f(x) = c\psi(x)$ can be multiplied by any arbitrary constant c , hence one can set $f^{(k)}(0) \rightarrow 0$ by letting $c \rightarrow 0$. Equation (10) in theorem 2 can be written as

$$L[x^{(n)} f(x)] = (-1)^n \{L[f(x)]\}^{(n)} \quad (12)$$

Similarly the inverse Laplace transformation

$$L^{-1}[s^{(n)} F(s)] = \{L^{-1}[f(x)]\}^{(n)} = f^{(n)}(x). \quad (13)$$

From these two theorems one can see that after the Laplace transform an m th order differential equation can become an n th order differential equation, where n does not depend on the original order m , but depends on the highest order of x multiplies $f^{(k)}(x)$, $k = 0,1,2, \dots, m$. If the highest order of x is 1, the differential equation after the Laplace transform is of first order and can be solved without much effort.

3. SOLVING SCHRÖDINGER EQUATION IN MOMENTUM SPACE BY USING LAPLACE TRANSFORMATION.

From the equation (5), it can be written by this way

$$\frac{d^2}{dp^2} \psi(p) - \frac{p^2}{\hbar^2 m^2 \omega^2} \psi(p) = -\frac{2E}{\hbar^2 m \omega^2} \psi(p) \quad (14)$$

It is convenient to work with a new variable (η) and a new parameter (λ) defined by

$$C = (\hbar^2 m^2 \omega^2)^{\frac{1}{4}} \quad \lambda = \frac{2EC^2}{\hbar^2 m \omega^2} \quad (15)$$

And the variable $\mathcal{E}$ with

$$p = C\mathcal{E} \quad (16)$$

(You can check the units of $(\hbar^2 m^2 \omega^2)^{\frac{1}{4}}$ to see they are the units of momentum, so $\mathcal{E}$ is indeed dimensionless here.)

Then equation (14) becomes.

$$\frac{d^2}{d\mathcal{E}^2} \psi(\mathcal{E}) + (\lambda - \mathcal{E}^2) \psi(\mathcal{E}) = 0 \quad (17)$$

We shall investigate first the solution of equation (17), when $\mathcal{E} \rightarrow \infty$. When $\mathcal{E}$ is very large $\lambda - \mathcal{E}^2 \cong -\mathcal{E}^2$ and equation (17) becomes

$$\frac{d^2}{d\mathcal{E}^2} \psi(\mathcal{E}) - \mathcal{E}^2 \psi(\mathcal{E}) = 0 \quad (18)$$

Its asymptotic solution is.

$$\Psi(\mathcal{E}) = e^{\mp \frac{\mathcal{E}^2}{2}} \quad (19)$$

Since substitution of equation (19) in equation (18) gives

$$(\mathcal{E}^2 + 1) e^{\mp \frac{\mathcal{E}^2}{2}} = \mathcal{E}^2 \Psi(\mathcal{E}) \quad (20)$$

Out of the two asymptotic solutions, $e^{\mp \frac{\mathcal{E}^2}{2}}$ is not acceptable as it diverges when $|\mathcal{E}| \rightarrow \infty$. The exact solution of equation (18) may be written as

$$\Psi(\mathcal{E}) = e^{-\frac{\mathcal{E}^2}{2}} H(\mathcal{E}) \quad (21)$$

Substitution equation (21) into equation (18), it becomes

$$v(\mathcal{E})'' - 2\mathcal{E} v(\mathcal{E})' + (\lambda - 1)v(\mathcal{E}) = 0 \quad (22)$$

Traditionally, after factoring out the asymptotic solution one uses the power expansion method to solve the equation. Here we use the Laplace transform instead. The method of factoring out the asymptotic solution reduces the highest order of the coefficients to one (the coefficients contain now only $\mathcal{E}$ and constants). After the Laplace transform the equation becomes a first-order differential equation. $L[v(\mathcal{E})] = V(s)$. From equation (2.7) one has

$$L[\mathcal{E} v'] = -(L[v'])' = -[sV(s)]' = -sV'(s) - V(s) \quad (23)$$

This implies

$$v(s) = C s^{-\frac{\lambda+1}{2}} e^{-\frac{s^2}{4}} \quad (24)$$

The invers Laplace transform yields

$$v(\mathcal{E}) = C \int s^{-\frac{(\lambda+1)}{2}} e^{-\frac{s^2}{4}} e^{s\mathcal{E}} ds \quad (25)$$

Note that in the following we shall denote all the constants by C , although they may represent different values in different places. Only when $(\lambda + 1) / 2 = n + 1$ and $n = 0, 1, 2, 3, \dots$, the residue in equation (25) is nonzero and one has.

$$v = CH_n(\mathcal{E}) \quad (26)$$

The integral form of the Hermite function is,

$$H_n(\mathcal{E}) = C \int s^{-(n+1)} e^{-\frac{s^2}{4}} e^{s\mathcal{E}} ds \quad (27)$$

The solution of ψ is then, by equations (21) and (26), the Hermite Gaussian function

$$\Psi(\mathcal{E}) = C e^{-\frac{\mathcal{E}^2}{2}} H(\mathcal{E}) \quad (28)$$

By substitution equation (17) and equation (15) in equation (28)

$$\Psi(p) = C e^{-\frac{p^2}{2\hbar m \omega}} H_n\left(\frac{p}{\sqrt{\hbar m \omega}}\right) \quad (29)$$

The energy spectra is, by equation (3.2) and $(\lambda + 1) / 2 = n + 1$,

$$E_n = \left(n + \frac{1}{2}\right) \hbar \omega \quad (30)$$

Normalization

The normalization condition is,

$$C^2 \int_{-\infty}^{\infty} \Psi^*(p) \Psi(p) dp = 1 \quad (31)$$

By substituting equation (29) in equation (31)

$$C^2 \int_{-\infty}^{\infty} e^{-\frac{p^2}{2\hbar m \omega}} H_n\left(\frac{p}{\sqrt{\hbar m \omega}}\right) \times e^{-\frac{p^2}{2\hbar m \omega}} H_n\left(\frac{p}{\sqrt{\hbar m \omega}}\right) dp = 1 \quad (32)$$

Or

$$C^2 \int_{-\infty}^{\infty} e^{-\frac{p^2}{\hbar m \omega}} \left[H_n \left(\frac{p}{\sqrt{\hbar m \omega}} \right) \right]^2 dp = 1 \quad (33)$$

By using the Generating function of Hermite polynomial of degree n ,

$$\int_{-\infty}^{\infty} e^{-y^2} [H_n(y)]^2 dy = 2^n n! \sqrt{\pi} \quad (34)$$

The constant of C becomes,

$$C = \left[\frac{1}{(\pi \hbar m \omega)^{\frac{1}{2}} 2^n n!} \right]^{\frac{1}{2}} \quad (35)$$

The normalized eigenfunction is given by

$$\Psi_n(p) = \left[\frac{1}{(\pi \hbar m \omega)^{\frac{1}{2}} 2^n n!} \right]^{\frac{1}{2}} e^{-\frac{p^2}{2\hbar m \omega}} H_n \left(\frac{p}{\sqrt{\hbar m \omega}} \right) \quad (36)$$

The wave function of ground state $\Psi_0(p)$ and the probability density $|\Psi_0(p)|^2$ are also illustrated in Figer (1)

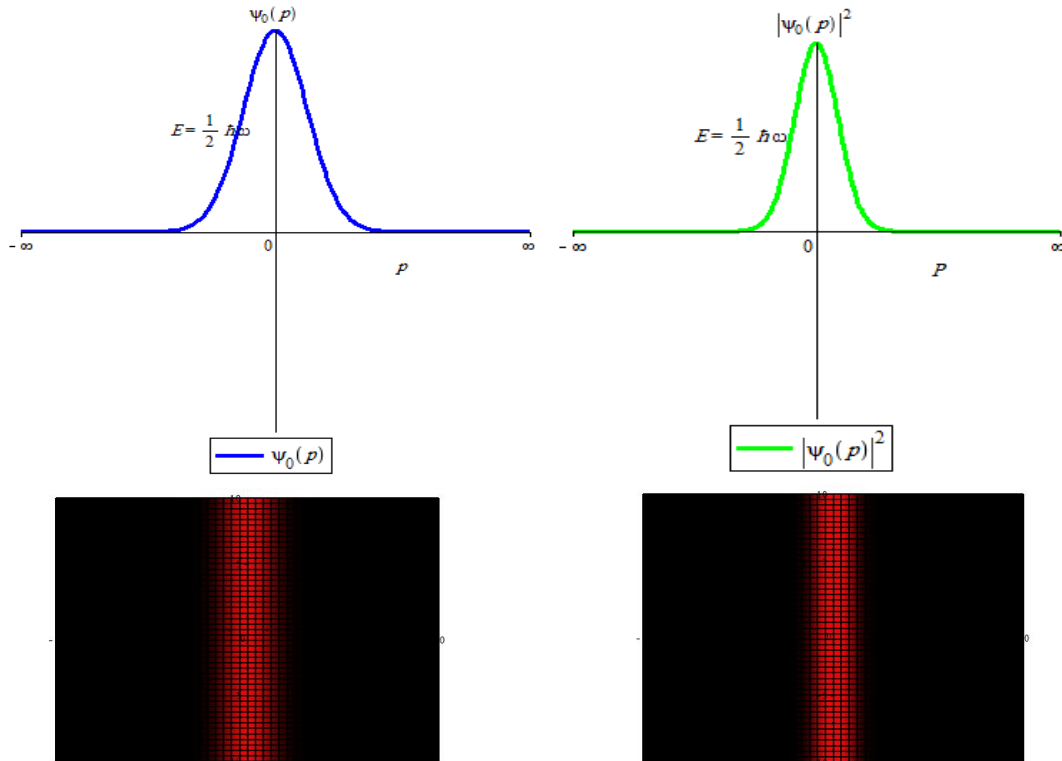


Figure (1) (a) Energy level of the (E_0) and wave function $\Psi_0(p)$ of the ground state.(b) Probability density ($|\Psi_0(p)|^2$) of the ground state.

The wave function of first excitation state $\Psi_1(p)$ and the probability density $|\Psi_1(p)|^2$ are also illustrated in Figer (2)

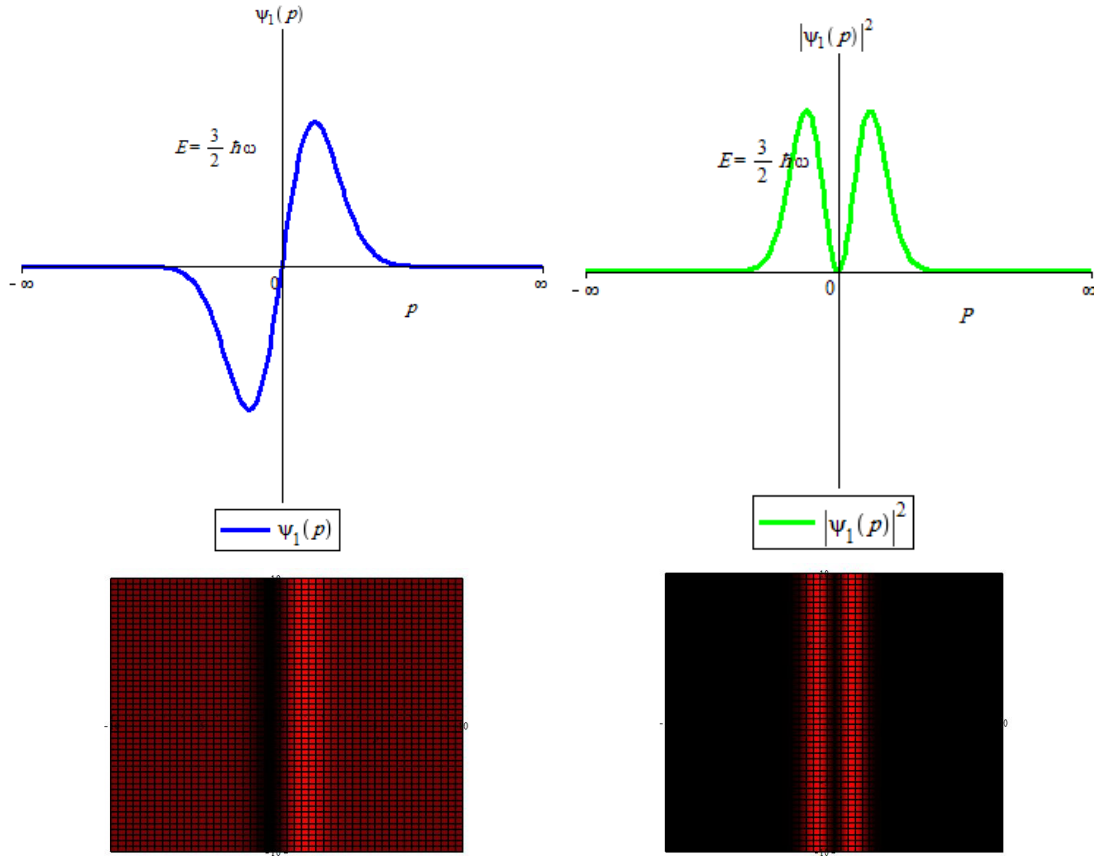


Figure (2) (a) Energy level of the (E_1) and wave function $\Psi_1(p)$ of the first excited state. (b) Probability density ($|\Psi_1(p)|^2$) of the first excited state.

4. CONVERT MOMENTUM REPRESENTATION TO POSITION REPRESENTATION BY USING FOURIER TRANSFORMATION.

The wave function of the ground state in momentum representation is,

$$\Psi_0(p) = \left[\frac{1}{\pi \hbar m \omega} \right]^{\frac{1}{4}} e^{-\frac{p^2}{2 \hbar m \omega}}$$

By using Fourier transformation

$$\Psi_0(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} e^{-\frac{ipx}{\hbar}} \Psi_0(p) dp \quad (37)$$

Or

$$\Psi_0(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} e^{-\frac{ipx}{\hbar}} \left[\frac{1}{\pi \hbar m \omega} \right]^{\frac{1}{4}} e^{-\frac{p^2}{2 \hbar m \omega}} dp \quad (38)$$

Change variable

$$y = \frac{p}{\sqrt{2\hbar m \omega}} + \sqrt{\frac{m\omega}{2\hbar}} ix \quad \text{and} \quad dy = \frac{1}{\sqrt{2\hbar m \omega}} dp \quad (39)$$

The equation (37) becomes,

$$\Psi_0(x) = \left[\frac{m\omega}{\pi \hbar} \right]^{\frac{1}{4}} e^{-\frac{m\omega}{2\hbar} x^2} \quad (40)$$

The equation of (40) is the wave function of the ground state of Harmonic oscillator in position representation.

5. POSITION-MOMENTUM UNCERTAINTY

The position of a plane wave is completely indeterminate as it is of infinite extent. Therefore, when waves are assigned to particles in motion indeterminacy arises automatically in the formalism because an electron wave of definite frequency is not localized. Heisenberg analyzed this indeterminacy proposed that no two canonically conjugate quantities can be measured simultaneously. For the canonically conjugate variable x and p mathematically, the principle is stated as [30].

$$\Delta x_p \Delta p = \frac{\hbar}{2} \quad (41)$$

The wave function of ground state in momentum representation is,

$$\Psi_0(p) = \left[\frac{1}{\pi \hbar m \omega} \right]^{\frac{1}{4}} e^{-\frac{p^2}{2 \hbar m \omega}}$$

$$(\Delta p)^2 = \langle p^2 \rangle - \langle p \rangle^2 \quad (42)$$

Expectation value of p ,

$$\langle p \rangle = \int_{-\infty}^{\infty} \left[\frac{1}{\pi \hbar m \omega} \right]^{\frac{1}{4}} e^{-\frac{p^2}{2 \hbar m \omega}} \times p \times \left[\frac{1}{\pi \hbar m \omega} \right]^{\frac{1}{4}} e^{-\frac{p^2}{2 \hbar m \omega}} dp = 0 \quad (43)$$

Expectation value of p^2 ,

$$\langle p^2 \rangle = \int_{-\infty}^{\infty} \left[\frac{1}{\pi \hbar m \omega} \right]^{\frac{1}{4}} e^{-\frac{p^2}{2 \hbar m \omega}} \times p^2 \times \left[\frac{1}{\pi \hbar m \omega} \right]^{\frac{1}{4}} e^{-\frac{p^2}{2 \hbar m \omega}} dp = \frac{1}{2} \hbar m \omega \quad (44)$$

By substituting the value of $\langle p \rangle$ and $\langle p^2 \rangle$ in equation (42),

$$(\Delta p)^2 = \frac{1}{2} \hbar m \omega \quad (45)$$

And

$$(\Delta x)^2 = \langle x^2 \rangle - \langle x \rangle^2 \quad (46)$$

Expectation value of x ,

$$\langle x \rangle = \int_{-\infty}^{\infty} \left[\frac{1}{\pi \hbar m \omega} \right]^{\frac{1}{4}} e^{-\frac{p^2}{2 \hbar m \omega}} \times (-i \hbar \frac{d}{dp}) \times \left[\frac{1}{\pi \hbar m \omega} \right]^{\frac{1}{4}} e^{-\frac{p^2}{2 \hbar m \omega}} dp = 0 \quad (47)$$

Expectation value of x^2 ,

$$\langle x^2 \rangle = \int_{-\infty}^{\infty} \left[\frac{1}{\pi \hbar m \omega} \right]^{\frac{1}{4}} e^{-\frac{p^2}{2 \hbar m \omega}} \times \left(-i \hbar \frac{d}{dp} \right)^2 \times \left[\frac{1}{\pi \hbar m \omega} \right]^{\frac{1}{4}} e^{-\frac{p^2}{2 \hbar m \omega}} dp = \frac{1}{2} \frac{\hbar}{m \omega} \quad (48)$$

By substituting value of $\langle x \rangle$ and $\langle x^2 \rangle$ in equation (46),

$$(\Delta x)^2 = \frac{1}{2} \frac{\hbar}{m \omega} \quad (49)$$

So

$$\Delta x_p \Delta p = \frac{\hbar}{2} \quad (50)$$

CONCLUSION

A new analytical technique has been introduced to obtain energy spectra and particle distributions for Harmonic oscillator in momentum space. The main advantage of this technique is that it provides a suitable way to calculate the differential equation with variable coefficient. Maple program has been used for solving Schrödinger equation and drawing graph for ground and 1st excited state. Fourier transformation has been used to convert the momentum representation wavefunction to the position representation wavefunction, we proved that this technique is successful and it can be used to solve different quantum systems for different potential energies.

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